Transient Couette Flow: Explicit and Implicit Formulations

~23BME013

Aim: To compare explicit FTCS and implicit backward-Euler solutions of transient viscous momentum diffusion against the analytical series solution.

Case definition and parameters: Simulations were analyzed on the following parameters.

Fourier number for Explicit calculation: 0.25, 0.45, 0.55
Fourier number for Implicit calculation: 5
Velocity: 1
t_end: 2
ny: 81
nu: 0.01

Governing Equation:

$$\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial y^2}, \qquad \nu \frac{\Delta t}{\Delta y^2} = Fo.$$

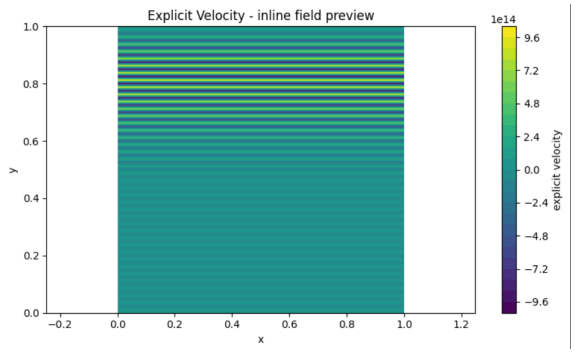
For FTCS, stability requirements Fo ≤ 0.5.

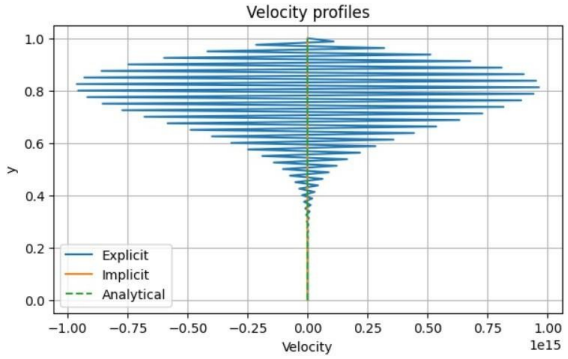
Boundary Conditions: Initially stationary fluid; lower plate fixed; upper plate impulsively set to speed U; no pressure gradient.

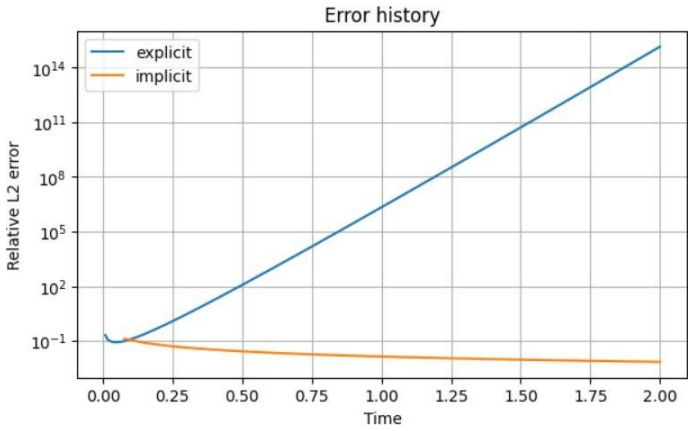
Numerical Method: Initially stationary fluid; lower plate fixed; upper plate impulsively set to speed U; no pressure gradient.

Observations: Initially stationary fluid; lower plate fixed; upper plate impulsively set to speed U; no pressure gradient. explicit Fourier numbers 0.25, 0.45, and 0.55. Explain the stability boundary and compare both schemes at the same physical time.

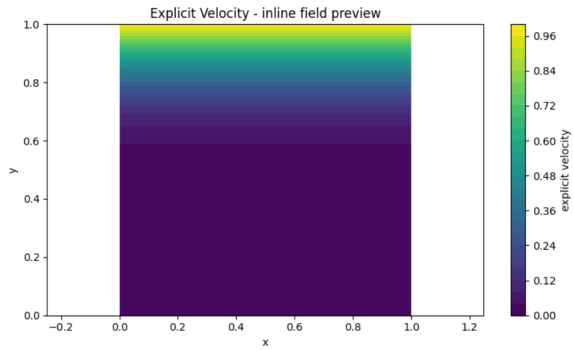
For explicit fourier number 0.55

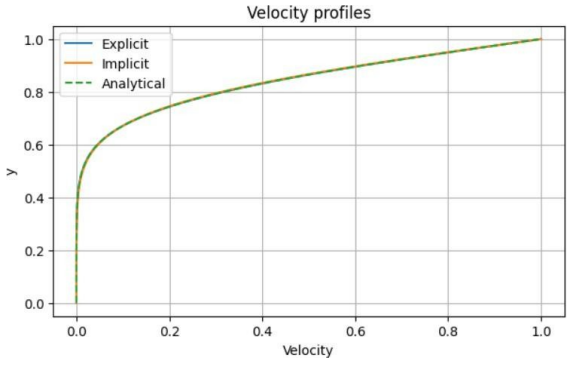


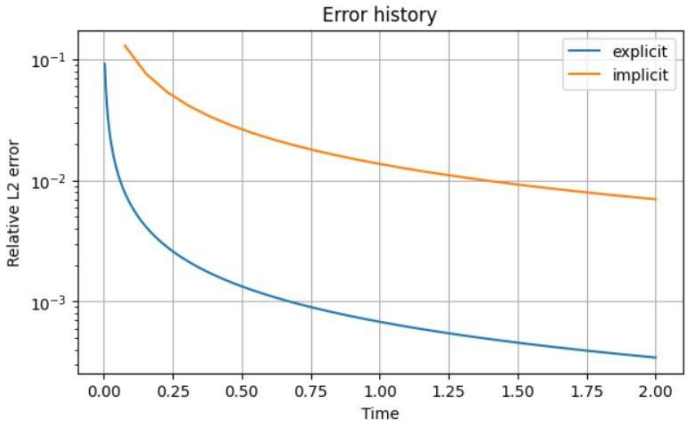




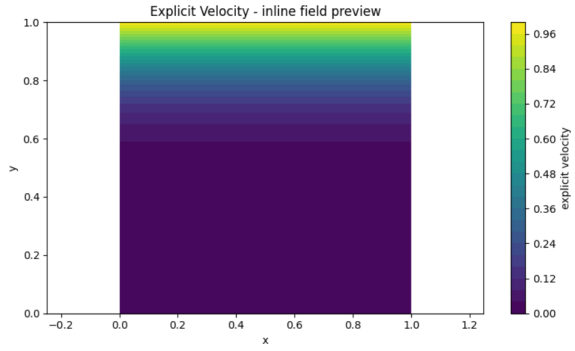
For explicit fourier number 0.45

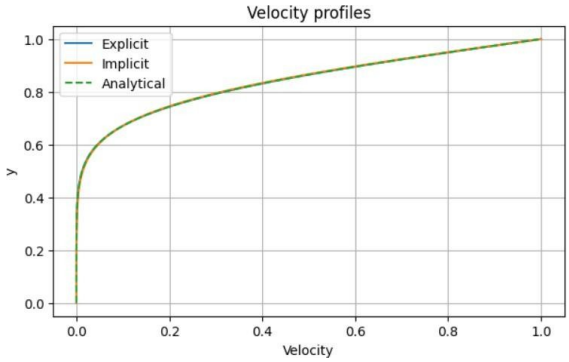


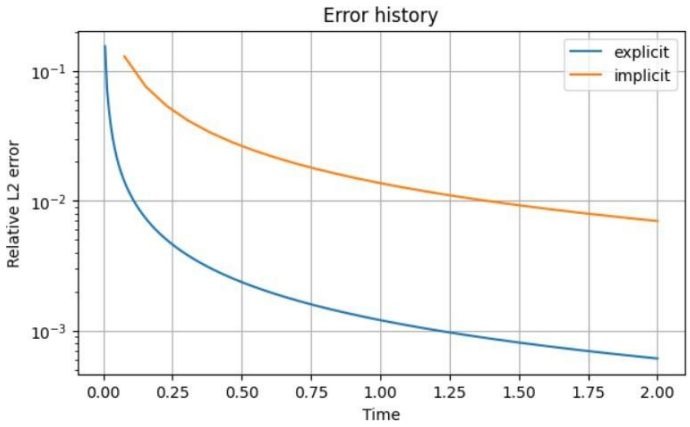




For explicit fourier number 0.25







Results:

Parameter	**Result**
Kinematic viscosity, nu	0.01
Upper plate velocity, U	1
Number of grid points, ny	81
Final simulation time, tend	2
Explicit Fourier number, Fo	0.25, 0.45, 0.55
Implicit Fourier number, Fo	5
Maximum stable Fourier number for FTCS	0.5
Relative L2 error at t=2, explicit Fo=0.25	6×10^{-4}
Relative L2 error at t=2, explicit Fo=0.45	3×10^{-4}
Relative L2 error at t=2, explicit Fo=0.55	10^{15}
Relative L2 error at t=2, implicit Fo=5	7×10^{-3}
Velocity profile for Fo=0.25 and 0.45	Closely matches analytical solution
Velocity profile for Fo=0.55	Unstable, with unbounded oscillations
Overall stability	Explicit stable for Fo ≤ 0.5; unstable for Fo > 0.5

Conclusion:

The transient Couette flow problem was solved using the explicit Forward-Time Central-Space method and the implicit backward-Euler method. The numerical solutions were compared with the analytical solution for viscous momentum diffusion between two parallel plates. The upper plate was impulsively set to a velocity of $U = 1$, while the lower plate remained stationary. The simulation used a kinematic viscosity of $\nu = 0.01$, 81 grid points in the wall-normal direction, and a final simulation time of $t = 2$.

Effect of Fourier number on explicit-scheme stability: The explicit FTCS formulation was tested at Fourier numbers of 0.25, 0.45, and 0.55. For the one-dimensional diffusion equation, the stability requirement is $Fo \leq 0.5$. At $Fo = 0.25$ and $Fo = 0.45$, the numerical solutions remained stable and closely followed the analytical velocity profile. Their relative L_2 errors at $t = 2$ were approximately 6×10^{-4} and 3×10^{-4}, respectively, as estimated from the error-history plots.

When the Fourier number was increased to 0.55, exceeding the theoretical stability limit, the explicit solution became unstable. The velocity field developed rapidly growing oscillations, with the plotted velocity magnitude reaching approximately 10^{15}. The relative L^2 error also increased dramatically, reaching the order of 10^{15} by the end of the simulation. This behaviour represents numerical divergence rather than a physically meaningful velocity distribution. It occurs because the explicit time integration no longer satisfies the stability condition required for the diffusion equation.

Comparison with the implicit backward-Euler scheme: The implicit method was tested at $Fo = 5$, substantially above the explicit stability limit. Its error history remained bounded and decreased over the simulated interval, reaching approximately 7×10^{-3} at $t = 2$. This demonstrates the improved stability of the implicit formulation for large time steps. However, unconditional stability does not guarantee the smallest numerical error; the implicit solution can still exhibit greater temporal discretisation error than a stable explicit solution.

Comparison with the analytical solution: For the stable explicit cases, the numerical velocity profiles closely overlapped the implicit and analytical profiles. The velocity increased from zero at the stationary lower plate to the imposed upper-plate velocity of approximately 1. This confirms that both methods can reproduce the expected transient momentum diffusion when applied under appropriate numerical conditions. In contrast, the $Fo = 0.55$ explicit case failed to reproduce the physical velocity distribution because of its instability.

Overall, the results demonstrate that the Fourier number is a critical parameter governing the stability of the explicit FTCS method. Selecting $Fo \leq 0.5$ maintained a bounded and accurate solution in the tested cases, whereas exceeding this limit caused exponential growth of numerical errors and divergence. The implicit backward-Euler scheme remained stable at $Fo = 5$, making it more suitable when larger time steps are required, although accuracy must still be assessed independently of stability.